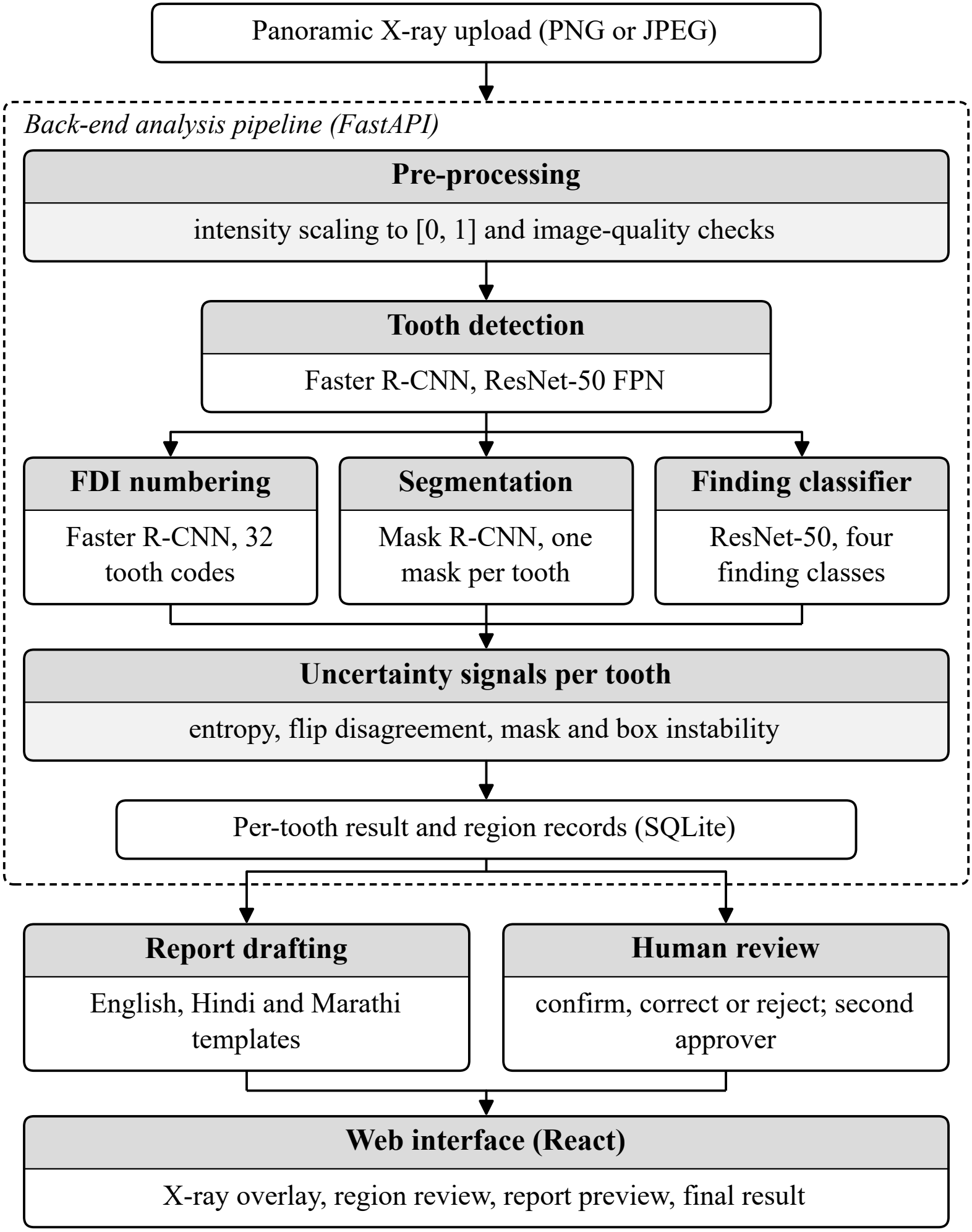

Panoramic X-ray upload (PNG or JPEG)
Back-end analysis pipeline (FastAPI)
Pre-processing
intensity scaling to [0, 1] and image-quality checks
Tooth detection
Faster R-CNN, ResNet-50 FPN
FDI numbering
Faster R-CNN, 32 tooth codes
Segmentation
Mask R-CNN, one mask per tooth
Finding classifier
ResNet-50, four finding classes
Uncertainty signals per tooth
entropy, flip disagreement, mask and box instability
Per-tooth result and region records (SQLite)
Report drafting
English, Hindi and Marathi templates
Human review
confirm, correct or reject; second approver
Web interface (React)
X-ray overlay, region review, report preview, final result